

SME Advisor

APC 2026 Competition Proposal

An AI-powered mobile advisory platform that helps Malaysian SME owners choose between cash payment, consumer BNPL, micro-credit, and government grants using machine learning and explainable decision support.

Repository: github.com/mizn08/SME-Advisor

1. Executive Summary

SME Advisor is a cross-platform mobile application backed by a REST API and PostgreSQL database. It addresses a critical gap faced by Malaysian small and medium enterprise (SME) owners: choosing the most capital-efficient way to fund business purchases among cash, Buy Now Pay Later (BNPL) products, micro-credit lines, and government financial aid schemes.

The system ingests SME transaction data via CSV upload, computes financial health KPIs, and uses an ensemble machine learning model (Logistic Regression, Random Forest, XGBoost) combined with a rule-based decision engine to recommend the optimal financing option. SHAP-style explanations provide transparency for each recommendation.

2. Problem Statement

Malaysian SMEs increasingly encounter diverse financing channels: consumer BNPL (Atome, Grab PayLater, Shopee), micro-credit from agencies (TEKUN, MARA, Agrobank, JHEV), B2B credit lines, and digitalisation grants (Geran Digital PMKS MADANI, MDEC). Owners lack a unified tool to compare total cost, cash-flow impact, and eligibility.

- Fragmented products with different interest structures and tenures

- Limited

3. Project Objectives

- Provide a mobile dashboard of SME financial health KPIs

- Simulat

4. Proposed Solution

SME Advisor delivers five integrated modules accessible via bottom navigation: Financial Health (dashboard), Purchase Simulator, AI Advisor (recommendation results), Government Aid Explorer, and Model Performance (transparency and history).

End-to-end flow: (1) Upload CSV transactions, (2) View KPIs and charts, (3) Enter purchase amount and category, (4) Receive ranked recommendation with cost and cash preserved estimates, (5) Review history and model metrics.

5. System Architecture

Three-tier architecture:

- Presentation: Flutter mobile app (Android APK, optional Web/Windows)

- Application

Key API endpoints:

- POST /upload-csv - ingest and clean transaction data

- GET /sn

6. Machine Learning Methodology

Training data comprises synthetic SME scenarios (2,200+ labelled samples) where label = 1 if external financing improves cash-flow stress. Features include days cash on hand, current ratio, burn rate, purchase amount, purchase-to-burn ratio, digitalisation flag, and industry indicators.

Three classifiers are trained and ensembled by averaging predicted probabilities. If probability < 0.5, Cash is recommended. Otherwise a rule engine compares BNPL offers, micro-credit lines, and eligible grants, prioritising digitalisation grants for qualifying purchases. XGBoost SHAP values (or heuristic fallback) explain top three drivers in the API and mobile UI.

7. Government & Financing Integration

Pre-seeded schemes include:

- BNPL: Atome Pay in 3, Grab PayLater, Shopee SPayLater

- Micro-cr

8. Technology Stack

- Mobile: Flutter, Provider, Dio, fl_chart, SharedPreferences, file_picker

- Backend

9. Innovation & Impact

- First mobile BNPL advisor integrating Malaysian government aid for SMEs

- Hybrid M

10. Implementation Status

- Fully functional backend API with Docker deployment

- Flutter m

11. Deployment & Demonstration

Local and competition demo: docker compose up --build exposes API on port 8000. Public judging access via ngrok tunnel (https://*.ngrok-free.app/docs). Flutter APK configured with --dart-define=API_BASE for remote API connectivity.

12. Future Enhancements

- Cloud VM deployment with HTTPS and fixed domain

- Real ba

13. Conclusion

SME Advisor demonstrates a viable, explainable, and locally relevant financing advisory platform for Malaysian SMEs. By combining machine learning, structured financial data, and government scheme awareness in a mobile-first experience, the project supports smarter capital decisions and improved cash-flow resilience for the backbone of Malaysia's economy.